Quantum theory of infinite systems

from fields to information

Hans Halvorson

November 12, 2013

Elementary quantum mechanics

states: normalized vectors in a Hilbert space $\mathcal{H}$. More generally: trace class operators on $\mathcal{H}$.

observables: hermitian (self-adjoint) linear operators on $\mathcal{H}$.

statistical algorithm: the expectation value of O in state ρ is given by

$$\mathrm{Tr}(\rho O) = \sum_{i=1}^{n} \lambda_i \langle \psi_i, O\psi_i \rangle.$$

dynamics the Hamiltonian H gives rise to a one-parameter group $U_t = \exp(-itH)$ of unitary operators on $\mathcal{H}$.

Elementary quantum mechanics

- The linear operators $\mathcal{B}(\mathcal{H})$ on $\mathcal{H}$ form an **algebra** closed under matrix sum, matrix multiplication, scalar multiplication, and the adjoint (hermitian conjugation) operation.
- This algebra has a natural **norm**: $\|O\|$ is the supremum of the eigenvalues of O.
- The states (density operators) form a **convex set**. The pure states correspond to unit vectors in $\mathcal{H}$.
- The states (density operators) correspond to positive, linear maps from $\mathcal{B}(\mathcal{H})$ to $\mathbb{C}$.

C^*-algebras

Definition: An abstract C^*-algebra $\mathcal{A}$ is a set equipped with operations:

- sum
- product
- complex scalar multiplication
- adjunction
- a norm $O \mapsto \|O\|$.

C^*-algebras

Fact: Every finite-dimensional C^*-algebra is a direct sum of matrix algebras.
But which C^*-algebras are not matrix algebras? And why might we need them for quantum theory?

states and probabilities

A **state** on $\mathcal{A}$ is a normalized, positive, linear map from $\mathcal{A}$ to $\mathbb{C}$.

The **transition probability** is defined by

$$p(\omega, \rho) \;=\; 1 - \frac{1}{4}\|\omega - \rho\|^2,$$

where

$$\|\omega - \rho\| \;=\; \sup\{|\omega(A) - \rho(A)| : A \in \mathcal{A}, \|A\| \leq 1\}.$$

Example: infinite dimensional Hilbert spaces

Let $\mathcal{H}$ be the linear space of square-integrable functions from $\mathbb{R}^n$ to $\mathbb{C}$. Then $\mathcal{H}$ is an infinite dimensional Hilbert space.

A linear operator $O : \mathcal{H} \to \mathcal{H}$ is said to be *bounded* if there is a $r < \infty$ such that $\|O\psi\| < r$ for all unit-length ψ in $\mathcal{H}$. The bounded linear operators $\mathcal{B}(\mathcal{H})$ on $\mathcal{H}$ form a C^*-algebra.

Example: infinite spin chain

Objective: Form an infinite tensor product of the matrix algebra M

observables: the algebra $\mathcal{A} = \bigotimes_{i\in\mathbb{Z}} M_i$ is generated by operators

$$\cdots \otimes A_{-2} \otimes A_{-1} \otimes A_0 \otimes A_1 \otimes A_2 \otimes \cdots$$

where $A_i = I$ for all but a finite number of entries.

states: every vector $\otimes_{i\in\mathbb{Z}}\psi_i$ induces a state on $\mathcal{A}$.

But there are too many states! The Hilbert space would have uncountably infinite dimension. And there are too few observables!

Representations of C^*-algebras

A $*$-**homomorphism** is a mapping $\pi : \mathcal{A} \to \mathcal{B}$ that preserves sum, product, and adjunction.

A **representation** of $\mathcal{A}$ on $\mathcal{H}$ is a $*$-homomorphism of $\mathcal{A}$ into $\mathcal{B}(\mathcal{H})$.

A $*$-homomorphism $\pi : \mathcal{A} \to \mathcal{B}$ gives rise to a mapping $\pi^* : S(\mathcal{B}) \to S(\mathcal{A})$.

$$\mathcal{A} \longrightarrow \mathcal{B} \longrightarrow \mathbb{C}$$

Representations of C^*-algebras

Two representations $(\mathcal{H}, \pi)$ and $(\mathcal{K}, \phi)$ of $\mathcal{A}$ are said to be **unitarily equivalent** just in case there is a unitary mapping $U : \mathcal{H} \to \mathcal{K}$ such that

$$U\pi(A) \;=\; \phi(A)U, \qquad \forall A \in \mathcal{A}.$$

The GNS representation theorem

For each state ω of $\mathcal{A}$, there is a representation $(\mathcal{H}, \pi)$ of $\mathcal{A}$ and a unit vector $\Omega \in \mathcal{H}$ such that

$$\omega(A) \;=\; \langle \Omega, \pi(A)\Omega \rangle, \qquad\qquad \forall A \in \mathcal{A},$$

and the vectors $\{A\Omega : A \in \mathcal{A}\}$ are dense in $\mathcal{H}$.

Von Neumann algebras

A **von Neumann algebra** $\mathcal{M}$ is a $*$-closed algebra acting on a Hilbert space $\mathcal{H}$ and such that $\mathcal{M} = \mathcal{M}''$, where

$$\mathcal{M}' = \{X \in \mathcal{B}(\mathcal{H}) : [X, A] = 0, \forall A \in \mathcal{M}\}.$$

Fact: Every von Neumann algebra is a C^*-algebra.

Fact: If $\mathcal{A}$ is a C^*-algebra acting concretely on $\mathcal{H}$, then $\mathcal{A}''$ is the smallest von Neumann algebra containing $\mathcal{A}$.

Example: Every matrix algebra is a von Neumann algebra. More generally, $\mathcal{B}(\mathcal{H})$ is a von Neumann algebra.

classification of von Neumann algebras

	type I	type II	type III
application	elementary QM	QSM	QFT

Each unitary operator $U \in \mathcal{A}$ induces an inner automorphism $\mathrm{ad}_U : \mathcal{A} \to \mathcal{A}$ defined by

$$\mathrm{ad}_U(A) = U^* A U, \qquad \forall A \in \mathcal{A}.$$

A **covariant representation** of $(\mathcal{A}, t \mapsto \alpha_t)$ consists of a Hilbert space $\mathcal{H}$, a $*$-homomorphism $\pi : \mathcal{A} \to \mathcal{B}(\mathcal{H})$, and a continuous representation $t \mapsto U_t$ of $\mathbb{R}$ on $\mathcal{H}$ such that

$$\pi(\alpha_t(A)) \;=\; U_t^* \pi(A) U_t, \qquad \forall A \in \mathcal{A}.$$

Dynamical GNS theorem: If ω is an invariant state of $(\mathcal{A}, t \mapsto \alpha_t)$, then the GNS representation is covariant, and the GNS vector Ω is invariant under U_t.

A linear map $T : \mathcal{A} \to \mathcal{B}$ is said to be **positive** just in case $T(A) \geq 0$ for all $A \geq 0$, and normalized if $T(I) = I$.

A linear map $T : \mathcal{A} \to \mathcal{B}$ is said to be **completely positive** just in case for each $n \in \mathbb{N}$ the map $T \otimes I_n : \mathcal{A} \otimes M_n(\mathbb{C}) \to \mathcal{A} \otimes M_n(\mathbb{C})$ is positive.

A map $T : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ is CP just in case there is a Hilbert space $\mathcal{K}$ and a unitary map $U : \mathcal{H} \otimes \mathcal{K} \to \mathcal{H} \otimes \mathcal{K}$ such that

$$T(A) \;=\; \mathrm{Tr}(U^*(A \otimes I)U).$$

the SNAG theorem

Let $\mathbf{x} \to U(\mathbf{x})$ be a continuous representation of the translation group of $\mathbb{R}^{n+1}$ in the unitary operators on $\mathcal{H}$. Then there are $n+1$ self-adjoint operators $P_0, P_1, \ldots, P_n$ on $\mathcal{H}$ such that

$$U(\mathbf{x}) \;=\; \exp(-ix_0P_0)\exp(-ix_1P_1)\cdots\exp(-ix_nP_n).$$

The quadruple (P_0, P_1, P_2, P_3) is called the 4-momentum of the system. The operator P_0 generates time translation, and is called the Hamiltonian.